Objective

Dive deep into working with datasets using *pandas*.

Things To Learn

- Creating and working with *pandas dataframes* and *series*.
- Loading and getting an overview of data.
- Manipulating datasets.
- Selecting, grouping and sorting data.

Submission Guidelines

- Your finished *Jupyter Notebook* - both as `.ipynb` and exported `.pdf`.

Task: Manually Creating A *Dataframe*

Task: Import *pandas* and manually create a *dataframe* containing data about a domain **you're passionate about**. Your dataframe should...

- ...contain at least 4 rows.
- ...contain at least 3 column, of which at least one should be numeric and one should be suitable as a *key* (i.e. *index*).

Next, manually create a *series* and add it as additional column to your dataset.

Finally, set the index of your dataframe to a suitable column.

In [38]:
```
# Spitzer
import pandas as pd

data = {
    'Title': ['1984', 'To Kill a Mockingbird', 'The Great Gatsby', 'Pride an
    'Author': ['George Orwell', 'Harper Lee', 'F. Scott Fitzgerald', 'Jane A
    'Year': [1949, 1960, 1925, 1813],
    'Pages': [328, 376, 180, 432]
}
df_books = pd.DataFrame(data)

genre_series = pd.Series(['Dystopian', 'Fiction', 'Classic', 'Romance'], nam

df_books['Genre'] = genre_series
```

```python
df_books.set_index('Title', inplace=True)

df_books
```

Out[38]:

Title	Author	Year	Pages	Genre
1984	George Orwell	1949	328	Dystopian
To Kill a Mockingbird	Harper Lee	1960	376	Fiction
The Great Gatsby	F. Scott Fitzgerald	1925	180	Classic
Pride and Prejudice	Jane Austen	1813	432	Romance

Task: Importing And Getting An Overview

In [39]:

```python
# Spitzer
import pandas as pd

df = pd.read_csv('ramen-ratings.csv')

print("First 5 rows:")
print(df.head())
print("\nLast 5 rows:")
print(df.tail())
print("\nRandom 5 rows:")
print(df.sample(5))

print("\nSummary statistics:")
print(df.describe())

print("\nData types:")
print(df.dtypes)

df.set_index('Review #', inplace=True)
print("\nIndex set to Review #")
print(df.head())
```

```
First 5 rows:
   Review #           Brand  \
0      2580       New Touch   
1      2579        Just Way   
2      2578          Nissin   
3      2577         Wei Lih   
4      2576  Ching's Secret   

                                             Variety Style Country Stars  \
0                          T's Restaurant Tantanmen    Cup   Japan  3.75   
1  Noodles Spicy Hot Sesame Spicy Hot Sesame Guan...  Pack  Taiwan     1   
2                      Cup Noodles Chicken Vegetable   Cup     USA  2.25   
3                      GGE Ramen Snack Tomato Flavor  Pack  Taiwan  2.75   
4                                    Singapore Curry  Pack   India  3.75   

  Top Ten  
0     NaN  
1     NaN  
2     NaN  
3     NaN  
4     NaN  

Last 5 rows:
      Review #     Brand                                            Variety  
\
2575         5     Vifon  Hu Tiu Nam Vang ["Phnom Penh" style] Asian Sty...   
2576         4   Wai Wai                     Oriental Style Instant Noodles   
2577         3   Wai Wai                                     Tom Yum Shrimp   
2578         2   Wai Wai                               Tom Yum Chili Flavor   
2579         1  Westbrae                                         Miso Ramen   

     Style   Country Stars Top Ten  
2575  Bowl   Vietnam   3.5     NaN  
2576  Pack  Thailand     1     NaN  
2577  Pack  Thailand     2     NaN  
2578  Pack  Thailand     2     NaN  
2579  Pack       USA   0.5     NaN  

Random 5 rows:
      Review #             Brand                                 Variety Style  
\
104       2476     Samyang Foods                     Kimchi Stew Ramyun  Bowl   
1273      1307            Nissin                Cup Noodle Cheese Curry   Cup   
1430      1150  Asian Thai Foods  Krrish Instant Noodles Chicken Flavor  Pack   
660       1920            Itsuki      Ramen Tonkotudou Kumamoto Noodles  Pack   
297       2283            Nissin         Raoh Pork Bone Soy Soup Noodle  Pack   

          Country Stars Top Ten  
104   South Korea     4     NaN  
1273        Japan  3.75     NaN  
1430        Nepal   3.5     NaN  
660         Japan  0.75     NaN  
297         Japan     5     NaN  

Summary statistics:
          Review #
```

```
count  2580.000000
mean   1290.500000
std     744.926171
min       1.000000
25%     645.750000
50%    1290.500000
75%    1935.250000
max    2580.000000

Data types:
Review #     int64
Brand       object
Variety     object
Style       object
Country     object
Stars       object
Top Ten     object
dtype: object

Index set to Review #
                   Brand                                            Variety  \
Review #
2580           New Touch                          T's Restaurant Tantanmen
2579            Just Way  Noodles Spicy Hot Sesame Spicy Hot Sesame Guan...
2578              Nissin                      Cup Noodles Chicken Vegetable
2577             Wei Lih                      GGE Ramen Snack Tomato Flavor
2576      Ching's Secret                                    Singapore Curry

         Style Country Stars Top Ten
Review #
2580       Cup   Japan  3.75     NaN
2579      Pack  Taiwan     1     NaN
2578       Cup     USA  2.25     NaN
2577      Pack  Taiwan  2.75     NaN
2576      Pack   India  3.75     NaN
```

Read the *ramen* ratings from the provided file and get an overview of the data:

- Take a look at some rows from the top, bottom or random positions...
- Use summary functions to get a statistical overview of the data.
- Find out what data types we're dealing with.
- Identify what column would be suitable as index and set it!

In [40]:

```python
# Spitzer
df = pd.read_csv('ramen-ratings.csv')

print("First 5 rows:")
print(df.head())
print("\nLast 5 rows:")
print(df.tail())
print("\nRandom sample of 5 rows:")
print(df.sample(5))

print("\nStatistical summary:")
```

```
print(df.describe())

print("\nData types:")
print(df.dtypes)
print("\nDataFrame info:")
print(df.info())

df = df.set_index('Review #')
print("\nDataFrame after setting index:")
print(df.head())
```

```
First 5 rows:
   Review #           Brand  \
0      2580       New Touch
1      2579        Just Way
2      2578          Nissin
3      2577         Wei Lih
4      2576  Ching's Secret

                                             Variety Style Country Stars  \
0                          T's Restaurant Tantanmen    Cup   Japan  3.75
1  Noodles Spicy Hot Sesame Spicy Hot Sesame Guan...  Pack  Taiwan     1
2                      Cup Noodles Chicken Vegetable   Cup     USA  2.25
3                      GGE Ramen Snack Tomato Flavor  Pack  Taiwan  2.75
4                                    Singapore Curry  Pack   India  3.75

  Top Ten
0     NaN
1     NaN
2     NaN
3     NaN
4     NaN

Last 5 rows:
      Review #     Brand                                            Variety
\
2575         5     Vifon  Hu Tiu Nam Vang ["Phnom Penh" style] Asian Sty...
2576         4   Wai Wai                      Oriental Style Instant Noodles
2577         3   Wai Wai                                      Tom Yum Shrimp
2578         2   Wai Wai                                Tom Yum Chili Flavor
2579         1  Westbrae                                          Miso Ramen

     Style   Country Stars Top Ten
2575  Bowl   Vietnam   3.5     NaN
2576  Pack  Thailand     1     NaN
2577  Pack  Thailand     2     NaN
2578  Pack  Thailand     2     NaN
2579  Pack       USA   0.5     NaN

Random sample of 5 rows:
      Review #      Brand                                            Variety
\
1868       712  Four Seas                                            Seaweed
628       1952    Ve Wong  Kung-Fu Brand Instant Oriental Noodle Soup Art...
2307       273    Wai Wai                                 Minced Pork Tom Yum
1262      1318     Nissin                             Spa Oh Tarako Spaghetti
2230       350     Nissin                      Souper Meal Tomato Garlic Shrimp

     Style    Country Stars Top Ten
1868  Bowl  Hong Kong  3.25     NaN
628   Pack     Taiwan  3.75     NaN
2307  Pack   Thailand   2.5     NaN
1262  Bowl      Japan     4     NaN
2230  Bowl        USA   3.5     NaN

Statistical summary:
          Review #
```

```
count  2580.000000
mean   1290.500000
std     744.926171
min       1.000000
25%     645.750000
50%    1290.500000
75%    1935.250000
max    2580.000000

Data types:
Review #     int64
Brand       object
Variety     object
Style       object
Country     object
Stars       object
Top Ten     object
dtype: object

DataFrame info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2580 entries, 0 to 2579
Data columns (total 7 columns):
 #   Column    Non-Null Count  Dtype 
---  ------    --------------  ----- 
 0   Review #  2580 non-null   int64 
 1   Brand     2580 non-null   object
 2   Variety   2580 non-null   object
 3   Style     2578 non-null   object
 4   Country   2580 non-null   object
 5   Stars     2580 non-null   object
 6   Top Ten   41 non-null     object
dtypes: int64(1), object(6)
memory usage: 141.2+ KB
None

DataFrame after setting index:
                   Brand                                            Variety  
\
Review #                                                                      
2580           New Touch                          T's Restaurant Tantanmen  
2579            Just Way  Noodles Spicy Hot Sesame Spicy Hot Sesame Guan...
2578              Nissin                      Cup Noodles Chicken Vegetable
2577             Wei Lih                      GGE Ramen Snack Tomato Flavor
2576      Ching's Secret                                    Singapore Curry

         Style Country Stars Top Ten
Review #                            
2580       Cup   Japan  3.75     NaN
2579      Pack  Taiwan     1     NaN
2578       Cup     USA  2.25     NaN
2577      Pack  Taiwan  2.75     NaN
2576      Pack   India  3.75     NaN
```

Task: Dealing With Erroneous Rows

As you surely noticed, the star ratings are not included in the numerical statistics. Find out why and fix this, so we can work with the column:

- Take a look at the column's data type (should have been determined in the last task). Try casting it to `float` !
- Find out all the unique values in the column.
- Now that you know the faulty values, identify their rows and drop them. This should affect 3 rows.
- Try casting the the column again and make sure we can now calculate statistics on it by printing the mean of it.

In [41]:
```python
# Spitzer
print("Data type of Stars:", df['Stars'].dtype)

try:
    df['Stars'] = df['Stars'].astype(float)
except ValueError as e:
    print("Error casting to float:", e)

print("Unique values in Stars:")
print(df['Stars'].unique())

faulty_rows = df[~df['Stars'].str.replace('.', '').str.isnumeric()]
print("Faulty rows:")
print(faulty_rows)

df = df.drop(faulty_rows.index)

df['Stars'] = df['Stars'].astype(float)

print("Mean of Stars:", df['Stars'].mean())
```

```
Data type of Stars: object
Error casting to float: could not convert string to float: 'Unrated'
Unique values in Stars:
['3.75' '1' '2.25' '2.75' '4.75' '4' '0.25' '2.5' '5' '4.25' '4.5' '3.5'
 'Unrated' '1.5' '3.25' '2' '0' '3' '0.5' '4.00' '5.0' '3.50' '3.8' '4.3'
 '2.3' '5.00' '3.3' '4.0' '3.00' '1.75' '3.0' '4.50' '0.75' '1.25' '1.1'
 '2.1' '0.9' '3.1' '4.125' '3.125' '2.125' '2.9' '0.1' '2.8' '3.7' '3.4'
 '3.6' '2.85' '3.2' '3.65' '1.8']
Faulty rows:
                  Brand                                Variety Style  \
Review #
2548             Ottogi  Plain Instant Noodle No Soup Included  Pack
2458      Samyang Foods                             Sari Ramen  Pack
1587            Mi E-Zee                        Plain Noodles  Pack

              Country    Stars Top Ten
Review #
2548      South Korea  Unrated     NaN
2458      South Korea  Unrated     NaN
1587         Malaysia  Unrated     NaN
Mean of Stars: 3.6546759798214974
```

Task: Missing Values

We don't want to deal with missing values in this example. Do the following:

- Find out how many values are missing in each column.
- One column should be missing 2 values. Identify the relevant rows and delete them.
- One column should be missing a lot of values. Delete the whole column.

In [42]:
```
# Spitzer
print("Missing values per column:")
print(df.isnull().sum())

df = df.dropna(thresh=len(df.columns) - 1)

df = df.drop(columns=['Top Ten'])

print("After dropping:")
print(df.isnull().sum())
```

```
Missing values per column:
Brand          0
Variety        0
Style          2
Country        0
Stars          0
Top Ten     2536
dtype: int64
After dropping:
Brand       0
Variety     0
Style       0
Country     0
Stars       0
dtype: int64
```

Task: Create A Price-Column

Unfortunately the ramen reviews don't contain a price column. Let's *fake* one with rough estimates:

- A *pack* of ramen should cost 0.79.
- A *bowl* of ramen should cost 1.79.
- A *cup* of ramen should cost 1.29.
- A *tray* of ramen should cost 2.19.
- All other types should cost 1.09.

Create a function calculating the price based on the style. Use `map` to create a series and add it as a new column to your dataframe.

In [43]:
```
# Spitzer
def get_price(style):
    if style == 'Pack':
        return 0.79
    elif style == 'Bowl':
        return 1.79
    elif style == 'Cup':
        return 1.29
    elif style == 'Tray':
        return 2.19
    else:
        return 1.09

df['Price'] = df['Style'].map(get_price)
print(df[['Style', 'Price']].head())
```

```
         Style  Price
Review #
2580       Cup   1.29
2579      Pack   0.79
2578       Cup   1.29
2577      Pack   0.79
2576      Pack   0.79
```

Task: From Stars To Points

Let's switch from a *star rating* to a *point rating* between 1 and 100.

- Calculate the points based on the stars, where 5 stars equal 100 points.
- Change the column name from *Stars* to *Points*.

In [44]:
```
# Spitzer
df = df.rename(columns={'Stars': 'Points'})
df['Points'] = df['Points'] * 20
print(df[['Points']].head())
```

```
          Points
Review #
2580        75.0
2579        20.0
2578        45.0
2577        55.0
2576        75.0
```

Task: Create A Recommendation-Column

Let's create a new column containing a textual recommendation:

- Ramen with points higher or equal than 90 points get either *Amazing value!* (for prices benath 1.3) or *Expensive but delicious!*.
- Other ramen with ratings above 80 should read *Must-Try!*.
- Other cheap ramen with a price below 1 should read *Budget choice!*.
- All other ramen should read *Why not?*.

Since you need multiple columns to calculate the recommendation, you need to use `apply` . Check how often each recommendation text appears afterwards!

In [45]:

```
# Spitzer
df = df.loc[:, ~df.columns.duplicated()]

def get_recommendation(row):
    points = row['Points']
    price = row['Price']
    if points >= 90:
        if price < 1.3:
            return 'Amazing value!'
        else:
            return 'Expensive but delicious!'
    elif points > 80:
        return 'Must-Try!'
    elif price < 1:
        return 'Budget choice!'
    else:
        return 'Why not?'

df['Recommendation'] = df.apply(get_recommendation, axis=1)
print(df['Recommendation'].value_counts())
```

```
Recommendation
Budget choice!              1080
Why not?                     761
Amazing value!               441
Must-Try!                    149
Expensive but delicious!     144
Name: count, dtype: int64
```

Task: Export Data

At this point it makes sense to backup our processed dataframe. Export the data into a file `ramen_processed.csv` into a subfolder `output` and compare your results to the provided solution.

In [46]:

```
# Spitzer
df.to_csv('output/ramen_processed.csv')
print("Data exported to output/ramen_processed.csv")
```

```
Data exported to output/ramen_processed.csv
```

Task: Selecting Data (*Integer-Based*)

You can now either use the dataset you've worked on so far or - if in doubt - load our `ramen_processed_solution.csv` for the next tasks.

Solve the following tasks to sharpen your selecting skills:

- Get the first two columns of the last ten rows.
- Get the 4^{th} column of the 15^{th} row.
- Get the second and the last column of the second last ten rows.
- Get everything but the first column of the 20^{th}, 30^{th}, 40^{th} and 50th row.
- Get every column of the 100^{th} up to (and including) the 200^{th} row.
- Get the third column of every row.

In [47]:

```
# Spitzer
print("First two columns of last ten rows:")
print(df.iloc[-10:, :2])

print("\n4th column of 15th row:")
print(df.iloc[14, 3])

print("\nSecond and last column of second last ten rows:")
print(df.iloc[-20:-10, [1, -1]])

print("\nEverything but first column of 20th, 30th, 40th, 50th rows:")
print(df.iloc[[19, 29, 39, 49], 1:])

print("\nEvery column of 100th to 200th rows:")
print(df.iloc[99:200])

print("\nThird column of every row:")
print(df.iloc[:, 2])
```

```
First two columns of last ten rows:
             Brand                                            Variety
Review #
10           Smack                                     Vegetable Beef
9            Sutah                                         Cup Noodle
8           Tung-I                  Chinese Beef Instant Rice Noodle
7          Ve Wong                                      Mushroom Pork
6            Vifon                                           Nam Vang
5            Vifon  Hu Tiu Nam Vang ["Phnom Penh" style] Asian Sty...
4          Wai Wai                    Oriental Style Instant Noodles
3          Wai Wai                                     Tom Yum Shrimp
2          Wai Wai                               Tom Yum Chili Flavor
1        Westbrae                                         Miso Ramen

4th column of 15th row:
Hong Kong

Second and last column of second last ten rows:
                                     Variety  Recommendation
Review #
20                  Neoguri (Seafood'n'Spicy)  Budget choice!
19                                Shin Ramyun  Budget choice!
18                      Jin Ramen (Hot Taste)  Budget choice!
17                             Pancit Palabok  Budget choice!
16                          Kalgug-Su (Spicy)  Budget choice!
15                         Pojangmacha U-dong  Budget choice!
14                                        Hot  Budget choice!
13                                  Chow Mein  Amazing value!
12                              Shrimp Flavor  Budget choice!
11        Chicken Flavor Instant Soup Noodle  Budget choice!

Everything but first column of 20th, 30th, 40th, 50th rows:
                                                Variety Style      Country  \
Review #
2561                                     Premium Gomtang  Pack  South Korea
2551           Signature Tom Yum Flavor Instant Noodles  Pack    Singapore
2540      GGE Noodle Snack Wheat Crackers Mexican Spicy  Pack       Taiwan
2530           Ppushu Ppushu Noodle Snack Honey Butter   Pack  South Korea

          Points  Price  Recommendation
Review #
2561        80.0   0.79  Budget choice!
2551        80.0   0.79  Budget choice!
2540        65.0   0.79  Budget choice!
2530        40.0   0.79  Budget choice!

Every column of 100th to 200th rows:
                  Brand                                            Variety  \
Review #
2480            Acecook                                    Pork Wantan Men
2479              E-Zee                     Instant Noodles Chicken Flavour
2478        Kiki Noodle                     Scallion Oil & Soy Sauce Noodle
2477        Kiki Noodle                        Sichuan Spices Flavor Noodle
2476      Samyang Foods                                 Kimchi Stew Ramyun
...                 ...                                                ...
2383             Nissin  Cup Noodles Beef Flavor Ramen Noodle Soup (New...
```

```
2382          Nongshim                              Seaweed Instant Noodle
2381            Nissin         Demae Iccho Seafood Flavour Instant Noodle
2380          Pulmuone                Non-Fried Ramyun With Spicy Beef Broth
2379            Nissin  Cup Noodles Hot & Spicy Shrimp Flavor Ramen No...

         Style      Country  Points  Price  Recommendation
Review #
2480      Bowl        Japan    85.0   1.79       Must-Try!
2479      Pack     Malaysia    80.0   0.79  Budget choice!
2478      Pack       Taiwan   100.0   0.79  Amazing value!
2477      Pack       Taiwan   100.0   0.79  Amazing value!
2476      Bowl  South Korea    80.0   1.79        Why not?
...        ...          ...     ...    ...             ...
2383       Cup          USA    70.0   1.29        Why not?
2382       Cup  South Korea    10.0   1.29        Why not?
2381       Cup    Hong Kong    95.0   1.29  Amazing value!
2380      Pack  South Korea    75.0   0.79  Budget choice!
2379       Cup          USA    50.0   1.29        Why not?

[101 rows x 7 columns]

Third column of every row:
Review #
2580     Cup
2579    Pack
2578     Cup
2577    Pack
2576    Pack
        ...
5       Bowl
4       Pack
3       Pack
2       Pack
1       Pack
Name: Style, Length: 2575, dtype: object
```

Task: Selecting Data (*Label-Based*)

Solve the following tasks to sharpen your selecting skills:

- Get the variety and textual recommendation for the review *#1235*.
- Get all columns but the textual recommendation for the reviews from (and including) *#5* to (and including) *#10*.
- Get the style and points for the reviews *#123*, *#234* and *#345*.
- Get the variety, style and country for all reviews.

In [48]:

```python
# Spitzer
print("Variety and recommendation for #1235:")
print(df.loc[[1235], ['Variety', 'Recommendation']])

print("\nAll columns but recommendation for #5 to #10:")
print(df.loc[5:10].drop(columns=['Recommendation'], errors='ignore'))
```

```python
print("\nStyle and points for #123, #234, #345:")
print(df.loc[[123, 234, 345], ['Style', 'Points']])

print("\nVariety, style and country for all:")
print(df[['Variety', 'Style', 'Country']])
```

```
Variety and recommendation for #1235:
                                                   Variety  Recommendation
Review #
1235      Demae Ramen Miso Tonkotsu Artificial Pork Flav...  Budget choice!

All columns but recommendation for #5 to #10:
Empty DataFrame
Columns: [Brand, Variety, Style, Country, Points, Price]
Index: []

Style and points for #123, #234, #345:
         Style  Points
Review #
123       Pack    74.0
234       Pack    75.0
345       Pack    70.0

Variety, style and country for all:
                                                   Variety Style   Country
Review #
2580                                T's Restaurant Tantanmen    Cup     Japan
2579      Noodles Spicy Hot Sesame Spicy Hot Sesame Guan...  Pack    Taiwan
2578                          Cup Noodles Chicken Vegetable   Cup       USA
2577                          GGE Ramen Snack Tomato Flavor  Pack    Taiwan
2576                                        Singapore Curry  Pack     India
...                                                     ...   ...       ...
5         Hu Tiu Nam Vang ["Phnom Penh" style] Asian Sty...  Bowl   Vietnam
4                            Oriental Style Instant Noodles  Pack  Thailand
3                                            Tom Yum Shrimp  Pack  Thailand
2                                       Tom Yum Chili Flavor  Pack  Thailand
1                                                Miso Ramen  Pack       USA

[2575 rows x 3 columns]
```

Task: Conditional Selection

Solve the following tasks to sharpen your selection skills:

- Get everything about ramen that comes in a bowl.
- Get variety, style and points of all ramen that comes from Germany.
- Get the columns from brand up to country of all ramen that comes in a cup and has a rating lower than 10 points.
- Get brand, variety and points of all ramen either produced by *Samyang* or having a rating over 95 points.
- Get everything but price and recommendation for all ramen containing *Hello Kitty*

in their variety field.

- Get everything up to country for all ramen from the brands *Knorr*, *Vifon* and *Yum Yum*.

In [49]:
```
# Spitzer
print("Ramen in bowl:")
print(df[df['Style'] == 'Bowl'])

print("\nVariety, style, points from Germany:")
print(df[df['Country'] == 'Germany'][['Variety', 'Style', 'Points']])

print("\nBrand to country for cup and points <10:")
print(df[(df['Style'] == 'Cup') & (df['Points'] < 10)].loc[:, 'Brand':'Count

print("\nBrand, variety, points for Samyang or points >95:")
print(df[(df['Brand'] == 'Samyang') | (df['Points'] > 95)][['Brand', 'Variet

print("\nEverything but price/recommendation for Hello Kitty:")
print(df[df['Variety'].str.contains('Hello Kitty')].drop(columns=['Price', '

print("\nUp to country for Knorr, Vifon, Yum Yum:")
print(df[df['Brand'].isin(['Knorr', 'Vifon', 'Yum Yum'])].loc[:, :'Country']
```

```
Ramen in bowl:
                     Brand                                          Variety  \
Review #
2567                Nissin                   Deka Buto Kimchi Pork Flavor
2555         Samyang Foods                      Song Song Kimchi Big Bowl
2553                Nissin             Hakata Ramen Noodle White Tonkotsu
2552               MyKuali       Penang White Curry Rice Vermicelli Soup
2547      Sichuan Guangyou                  Chongqing Spicy Hot Noodles
...                    ...                                              ...
43             Kim Ve Wong        Jaopai Series: Vegetarian Instant Noodles
41             Little Cook                               Pork Mustard Stem
39               Lucky Me!                           Supreme Bulalo Flavor
29                Mee Jang                                  Tom Yum Shrimp
5                    Vifon  Hu Tiu Nam Vang ["Phnom Penh" style] Asian Sty...

         Style      Country  Points  Price          Recommendation
Review #
2567      Bowl        Japan    90.0   1.79  Expensive but delicious!
2555      Bowl  South Korea    85.0   1.79                 Must-Try!
2553      Bowl        Japan    95.0   1.79  Expensive but delicious!
2552      Bowl     Malaysia   100.0   1.79  Expensive but delicious!
2547      Bowl        China    80.0   1.79                  Why not?
...        ...          ...     ...    ...                       ...
43        Bowl       Taiwan     0.0   1.79                  Why not?
41        Bowl     Thailand    10.0   1.79                  Why not?
39        Bowl  Philippines    60.0   1.79                  Why not?
29        Bowl     Thailand    70.0   1.79                  Why not?
5         Bowl      Vietnam    70.0   1.79                  Why not?

[481 rows x 7 columns]

Variety, style, points from Germany:
                                                Variety Style  Points
Review #
2498                            Demae Ramen Spicy Beef  Pack    75.0
2497                                 Cup Noodles Spicy   Cup    75.0
2496                                         Soba Thai  Pack    90.0
2495                        Cup Noodles Huhn (Chicken)   Cup    75.0
2494                         Demae Ramen Korean Kimchi  Pack    70.0
2493                                 Cup Noodles Curry   Cup    75.0
2491                             Soba Yakitori Chicken   Cup   100.0
2490                        Demae Ramen Garlic Chicken  Pack    70.0
2489                           Cup Noodles Ente (Duck)   Cup    60.0
2488                          Demae Ramen Thai Tom Yum  Pack    75.0
2487                                Soba Sukiyaki Beef   Cup    90.0
2486                                 Demae Ramen Spicy  Pack    70.0
2485                               Cup Noodles Shrimps   Cup    60.0
2484                       Demae Ramen Tokyo Soy Sauce  Pack    80.0
1975                        Soba Fried Noodles Classic  Pack    65.0
1916                       Soba Fried Noodles Teriyaki  Pack    60.0
1869                          Soba Fried Noodles Chili  Pack    70.0
1813                          Soba Fried Noodles Curry  Pack    85.0
1733                      Instant Noodles Beef Flavour  Pack    75.0
1637                 Instant Nudelsuppe Huhn Geschmack  Pack    60.0
1490            Soba Chili Noodles With Japanese Yakisoba Sauce   Cup    70.0
```

```
1476                      Instant Nudelsuppe Shrimp Flavour  Pack    70.0
1469            Oriental Style Instant Noodles Beef Flavour  Pack    65.0
1465                    Shrimp "Tom Yum" Instant Nudelsuppe  Pack    70.0
1368      Soba Teriyaki Noodles With Japanese Yakisoba S...   Cup    60.0
1222      Soba Classic Noodles With Japanese Yakisoba Sauce   Cup    75.0
1193        Soba Curry Noodles With Japanese Yakisoba Sauce   Cup    75.0

Brand to country for cup and points <10:
                    Brand                                            Variety
\
Review #
2426      Dr. McDougall's                        Vegan Pad Thai Noodle Soup
2023         Urban Noodle                  Authentic Street Food Black Bean
1956             Maruchan                          Spicy Tomato Salsa Ramen
1800       Crystal Noodle                         Soup All Natural Hot & Sour
1642           Campbell's                 Hearty Noodles Savoury Beef Flavour
1628                Azami                                 Kimchee Noodle Soup
1371           Campbell's                         Hearty Noodles Thai Flavour
584                Baijia  Single Noble Black Bone Chicken Sweet Potato T...
455                Baijia                  Single Noble Pickled Radish & Duck

         Style Country
Review #
2426       Cup     USA
2023       Cup      UK
1956       Cup   Japan
1800       Cup     USA
1642       Cup  Canada
1628       Cup  Canada
1371       Cup  Canada
584        Cup   China
455        Cup   China

Brand, variety, points for Samyang or points >95:
                    Brand                                            Variety
\
Review #
2570          Tao Kae Noi                       Creamy tom Yum Kung Flavour
2569             Yamachan                           Yokohama Tonkotsu Shoyu
2566               Nissin  Demae Ramen Bar Noodle Aka Tonkotsu Flavour In...
2563             Yamachan                                 Tokyo Shoyu Ramen
2559     Jackpot Teriyaki                                        Beef Ramen
...                   ...                                               ...
28                 Nissin                                      Chikin Ramen
16                Samyang                                 Kalgug-Su (Spicy)
15                Samyang                                Pojangmacha U-dong
14                Samyang                                               Hot
13         Sapporo Ichiban                                         Chow Mein

          Points
Review #
2570       100.0
2569       100.0
2566       100.0
2563       100.0
2559       100.0
```

```
...          ...
28         100.0
16          70.0
15          50.0
14          70.0
13         100.0

[405 rows x 3 columns]

Everything but price/recommendation for Hello Kitty:
                     Brand                                            Variety
\
Review #
2044                  Doll  Hello Kitty Dim Sum Noodle Japanese Soy Sauce ...
1369      A-Sha Dry Noodle               Hello Kitty Za Jiang (Soybean Sauce)
1328      A-Sha Dry Noodle                  Hello Kitty La Wei (Spicy Flavor)
1217                  Doll  Hello Kitty Dim Sum Noodle Japanese Curry Flavour
1190                Sanrio             Hello Kitty Hakata Shoyutonkotsu Ramen

         Style    Country  Points
Review #
2044       Cup  Hong Kong    60.0
1369      Pack     Taiwan    65.0
1328      Pack     Taiwan    75.0
1217       Cup  Hong Kong    70.0
1190      Bowl      Japan    75.0

Up to country for Knorr, Vifon, Yum Yum:
            Brand                                            Variety Style  \
Review #
2354        Vifon  Viet Cuisine Bun Rieu Cua Sour Crab Soup Insta...  Bowl
2346        Knorr                       Chatt Patta Instant Noodles  Pack
2193        Knorr                  Chicken Flavored Instant Noodles  Pack
1906        Vifon        Chicken Flavour Asian Style Instant Noodles  Pack
1867      Yum Yum   Premier Bowl Instant Noodles Stewed Pork Flavour  Bowl
1832      Yum Yum  Instant Flat Shaped Noodles Boat Noodles Nam T...  Pack
1819      Yum Yum          Premier Bowl Instant Noodles Suki Flavour  Bowl
1816        Vifon  Asian Style Instant Noodles Artificial Beef Fl...  Bowl
1749      Yum Yum  Jumbo Instant Noodles Tom Yum Kung Creamy Flavour  Pack
1715      Yum Yum                 Instant Noodles Minced Pork Flavor  Pack
1666        Vifon                  Curry Instant Noodle With Chicken   Cup
1522        Vifon                   Instant Porridge Seafood Flavour  Pack
1520      Yum Yum      Oriental Style Instant Noodles Wasabi Flavour  Pack
1510        Knorr                  Quick Serve Macaroni Wonton Broth  Pack
1437        Knorr     Quick Serve Macaroni Abalone & Chicken Flavour  Pack
1419        Vifon              Ngon-Ngon Tom Yum Minced Pork Noodle   Cup
1329        Vifon  Oriental Style Instant Noodle Mi Ga Chicken Fl...   Cup
1309      Yum Yum     Oriental Style Instant Noodles Seafood Flavour   Cup
1301        Knorr                  Quick Serve Macaroni Shrimp Broth  Pack
1277        Vifon  Mì Gà Tìm Chua Cay Hot & Sour Chicken Flavor I...  Pack
1156        Vifon  Oriental Style Instant Vermicelli Sour Crab Fl...  Pack
1050        Knorr    Japanese Pork Bone Flavour Quick Serve Macaroni  Pack
951         Vifon                        Pomidorowa (Mild Tomato)  Pack
873         Vifon                         Viet Rice Noodles Chicken  Pack
846         Vifon                                     Mi Kim Chee  Pack
831         Vifon                           Instant Porridge Chicken  Pack
```

716	Vifon	Hu Tieu Ca Stewed Fish	Bowl
620	Vifon	Phu Gia Bamboo Shoot Artificial Pork Bean Thread	Pack
589	Vifon	South Korean Style Kim Chee	Bowl
588	Vifon	Tu Quy Pork	Pack
583	Vifon	Phu Gia Cua Crab Bean Thread	Pack
563	Yum Yum	Duck	Pack
554	Yum Yum	Shrimp Tom Yum Flavor	Pack
546	Yum Yum	Vegetable	Pack
541	Yum Yum	Chicken	Pack
531	Yum Yum	Beef	Pack
504	Knorr	Pizza	Pack
467	Vifon	Tu Quy Sour Spicy Shrimp	Pack
426	Vifon	Tu quy Chicken	Pack
422	Vifon	Mi Lau Thai Thai Stle	Bowl
417	Vifon	Tu Quy Spicy Beef	Pack
406	Vifon	Asian Style Instant Noodles Beef	Pack
375	Vifon	Pho Bo Beef Rice Noodle	Pack
374	Vifon	Asian Style Instant Noodles Shrimp	Pack
368	Knorr	Chili Pork	Cup
330	Vifon	Phu Gia Instant Bean Thread Chicken	Pack
311	Vifon	Tasty Mushroom	Pack
228	Vifon	Pho Ga Instant rice Noodle	Bowl
177	Vifon	Hu Tieu Bo Kho An Lien Beef	Pack
163	Vifon	Asian Style Instant Noodles Chicken	Pack
127	Vifon	Mi Chay Vegetarian	Pack
6	Vifon	Nam Vang	Pack
5	Vifon	Hu Tiu Nam Vang ["Phnom Penh" style] Asian Sty...	Bowl

	Country
Review #	
2354	Vietnam
2346	Pakistan
2193	Pakistan
1906	Vietnam
1867	Thailand
1832	Thailand
1819	Thailand
1816	Vietnam
1749	Thailand
1715	Thailand
1666	Vietnam
1522	Vietnam
1520	Thailand
1510	Hong Kong
1437	Hong Kong
1419	Vietnam
1329	Vietnam
1309	Thailand
1301	Hong Kong
1277	Vietnam
1156	Vietnam
1050	Hong Kong
951	Poland
873	Vietnam
846	Vietnam
831	Vietnam

```
716          Vietnam
620          Vietnam
589          Vietnam
588          Vietnam
583          Vietnam
563         Thailand
554         Thailand
546         Thailand
541         Thailand
531         Thailand
504         Pakistan
467          Vietnam
426          Vietnam
422          Vietnam
417          Vietnam
406          Vietnam
375          Vietnam
374          Vietnam
368         Thailand
330          Vietnam
311          Vietnam
228          Vietnam
177          Vietnam
163          Vietnam
127          Vietnam
6            Vietnam
5            Vietnam
```

Task: Grouping And Sorting

Use grouping to solve the following tasks:

- Print out the number of reviews per brand in ascending order.
- Print out all the styles and their mean rating points - in descending order.
- Print out the minimum, maximum and mean price per brand, sorted by the maximum values.
- Print out all the values of the highest rated ramen per style!
- Print out the count and mean of ratings per style per brand, sorted by the count.

In [50]:
```python
# Spitzer
print("Number of reviews per brand (ascending):")
print(df.groupby('Brand').size().sort_values())

print("\nStyles and mean points (descending):")
print(df.groupby('Style')['Points'].mean().sort_values(ascending=False))

print("\nMin, max, mean price per brand (sorted by max):")
print(df.groupby('Brand')['Price'].agg(['min', 'max', 'mean']).sort_values('

print("\nHighest rated ramen per style:")
print(df.loc[df.groupby('Style')['Points'].idxmax()])
```

```
print("\nCount and mean per style per brand (sorted by count):")
print(df.groupby(['Style', 'Brand']).agg(count=('Points', 'size'), mean_poin
```

```
Number of reviews per brand (ascending):
Brand
1 To 3 Noodles      1
Weh Lih             1
Chikara             1
Westbrae            1
China Best          1
                 ...
Paldo              66
Mama               71
Maruchan           76
Nongshim           98
Nissin            381
Length: 355, dtype: int64

Styles and mean points (descending):
Style
Bar     100.000000
Box      85.833333
Pack     74.009162
Bowl     73.413721
Tray     70.902778
Can      70.000000
Cup      69.970000
Name: Points, dtype: float64

Min, max, mean price per brand (sorted by max):
                   min   max      mean
Brand
1 To 3 Noodles    0.79  0.79  0.790000
iNoodle           0.79  0.79  0.790000
Baixiang Noodles  0.79  0.79  0.790000
Baltix            0.79  0.79  0.790000
Binh Tay          0.79  0.79  0.790000
...                ...   ...       ...
Thai Kitchen      0.79  2.19  1.370000
Thai Pavilion     2.19  2.19  2.190000
Doll              0.79  2.19  0.940000
Vina Acecook      0.79  2.19  0.963529
Bon Go Jang       1.79  2.19  1.990000

[355 rows x 3 columns]

Highest rated ramen per style:
                           Brand  \
Review #
1155        Komforte Chockolates
2552                     MyKuali
2500      The Ramen Rater Select
2513                    Pringles
2558                        KOKA
2570                 Tao Kae Noi
2302                      Nissin

                                                    Variety Style    Country
\
```

```
Review #
1155                                        Savory Ramen    Bar        USA
2552             Penang White Curry Rice Vermicelli Soup  Bowl   Malaysia
2500                        Supreme Creamy Tom Yum Noodle   Box   Malaysia
2513        Nissin Top Ramen Chicken Flavor Potato Crisps    Can        USA
2558      Creamy Soup With Crushed Noodles Hot & Sour Fi...   Cup  Singapore
2570                          Creamy tom Yum Kung Flavour  Pack   Thailand
2302                                            Yakisoba  Tray      Japan

          Points  Price            Recommendation
Review #
1155       100.0   1.09            Amazing value!
2552       100.0   1.79  Expensive but delicious!
2500       100.0   1.09            Amazing value!
2513        70.0   1.09                  Why not?
2558       100.0   1.29            Amazing value!
2570       100.0   0.79            Amazing value!
2302       100.0   2.19  Expensive but delicious!

Count and mean per style per brand (sorted by count):
                            count  mean_points
Style Brand
Tray  Peyang                    1   100.000000
Bar   Komforte Chockolates      1   100.000000
Bowl  Haioreum                  1    70.000000
      Good Tto Leu Foods        1    80.000000
      Global Inspiration        1    45.000000
...                           ...          ...
Pack  Indomie                  47    81.914894
      Mama                     48    71.104167
Bowl  Nissin                   71    81.971831
Pack  Nissin                  121    79.049587
Cup   Nissin                  165    76.603030

[489 rows x 2 columns]
```

Good Job!